NGUYEN DUY TRANG

Open to Opportunities

Mid Mobile Engineer (Flutter, iOS, Android)

📍 Ho Chi Minh City, Vietnam • 🎂 2000 • 📞 (+84) 397 308 962 • ✉️ trangndps10349@gmail.com •
🐙 github.com/TrangDuyNguyen • 💼 linkedin.com/in/nguyen-duy-trang

PROFESSIONAL SUMMARY

OVERVIEW

Mobile Engineer with 5+ years of hands-on experience across Flutter, iOS, and Android. I have directly built and scaled social network platforms with over 1 million downloads, robust POS/Kiosk solutions across multiple industries (F&B, Retail, Spa), as well as Fintech applications (international money transfer, stock trading) and real-world aquaculture management systems. My core strengths lie in designing clean architectures, handling seamless offline data sync, and integrating peripheral hardware like thermal printers and barcode scanners. I am deeply focused on performance, maintaining consistent 60 FPS rendering and zero memory leaks. Recently, my focus is applying Multimodal Vision AI (Google Gemini 3.8 Flash) and Generative UI architectures in Flutter to build reactive, AI-driven mobile experiences.

VERIFIED PRODUCTION HIGHLIGHTS

METRICS

- **1M+ Downloads:** Operating large-scale social travel network & real-time OTT on iOS & Android.
- **-30% Checkout Time:** Optimized ordering, billing & checkout on multi-vertical POS & self-order Kiosks.
- **99.5%+ Crash-Free:** Production reliability, strict RAM lifecycle management, and locked 60 FPS.
- **Offline-First & RBAC:** SQLite order sync; permission boundaries for Cashier, Warehouse, Manager.
- **Realtime & Hardware:** Low-latency WebSocket chat/calls, ESC/POS thermal printers & barcode scanners.
- **277 Tests 100%:** Comprehensive automated test suite, Fastlane & GitHub Actions CI/CD pipelines.

TECHNICAL SKILLS MATRIX

COMPETENCIES

Languages & Core:	Flutter (Dart), Swift, SwiftUI, UIKit, Kotlin, TypeScript.
Architecture & State:	Clean Architecture, Riverpod 2.x, BLoC, Provider, MVVM, SOLID, Modularization, Offline-First, RBAC.
Real-Time, Media & Data:	WebSocket, Socket.IO, RESTful APIs, SQLite, Local Caching & Sync, iOS CallKit.
Hardware & POS Modules:	Platform Channels, Thermal POS (ESC/POS), Barcode Scanner, Warehouse Management, Multi-tier Pricing.
AI Tools, CI/CD & QA:	Cursor IDE, Claude Code, Fastlane, GitHub Actions, Bitrise, Git Worktrees, Scrum/Agile.

FEATURED FLAGSHIP PROJECT

PERSONAL ARCHITECTURE

ASTROBITE — AI Food Scanner & Voice Calorie Tracker / AstroVoice AI

2026 - Present

Senior Mobile Engineer & System Architect — Personal AI Mobile Project — Nutrition & Voice Logging

Tech Stack: Flutter 3.x, Dart 3.x, Riverpod 2.6 (Code-Gen, AsyncNotifier), AutoRoute 9.x, Google Gemini 3.8 Flash (Vision & NLU), on-device speech-to-text, iOS WidgetKit & Android AppWidget (home_widget), Apple Health & Health Connect, Fastlane, GitHub Actions CI/CD (277 Unit/Widget Tests).

- Architected AstroBite (v3.0.0) based on **Feature-First Clean Architecture** with Riverpod 2.x code-gen across 12 decoupled business modules.
- Engineered **AstroVoice AI** hands-free Vietnamese voice logging (`vi-VN`) with Gemini 3.8 Flash NLU parsing colloquial units and rendering GenUI 1-Tap Log in **~45ms** (~1.1s AI latency).
- Engineered camera scanner utilizing **Gemini Multimodal Vision AI** recognizing meals in **< 2.5s** with automatic broth/topping separation for Asian dishes (Pho, Broken Rice), slashing manual editing by 68% with **0 memory leaks**.
- Built **Flutter Generative UI (GenUI) Chat Cockpit** via Agent-to-UI (A2UI) protocol and Gemini 3.8 Flash, dynamically rendering interactive widgets (MealQuickLogCard, MacroBudgetGauge) with 1-tap Firestore logging.
- Developed **Cross-Platform OS Home Widgets** (iOS WidgetKit & Android AppWidget via home_widget) supporting instant camera scanner deep-link **< 300ms** and reactive data sync with zero background battery drain.
- Established a suite of **277 automated unit & widget tests (100% passing)** with zero flaky tests; automated multi-platform release builds and deployment using Fastlane & GitHub Actions.

NGUYEN DUY TRANG — COMMERCIAL WORK EXPERIENCE & CREDENTIALS

5+ YEARS HANDS-ON EXPERIENCE

COMMERCIAL WORK EXPERIENCE

TRACK RECORD

FINVIET TECHNOLOGY

08/2025 – 08/2026

Mobile Developer — Multi-Industry POS Kiosk Ecosystem (F&B, Retail & Spa Services)

Tech Stack: Flutter, Android Kotlin, BLoC, SQLite, Platform Channels, ESC/POS Thermal Printer, Barcode Scanner, RBAC, Fastlane.

- Architected **POS & self-order Kiosk** checkouts tailored for 3 business verticals: **F&B** (table layout, kitchen ticket printing), **Retail** (barcode scanning, rapid cart), and **Spa Services** (technician booking, treatment packages), cutting **30%** transaction time.
- Designed granular **Role-Based Access Control (RBAC)** across distinct operational roles (Cashier, Warehouse Keeper, Spa Technician, Store Manager, Admin).
- Developed **Goods & Multi-tier Pricing** module: custom pricing per branch, spa treatment packages, F&B food combos, and automatic promotional rules.
- Engineered **Warehouse Management** subsystem: stock-in/stock-out receipts, inter-branch transfers, and real-time inventory tracking.
- Built comprehensive **Reporting & Analytics**: shift revenue breakdown, multi-model sales statistics (F&B, Retail, Spa), and stock movement reconciliation.
- Architected **Offline-first** order sync (BLoC, SQLite) and developed custom **Platform Channels (Kotlin)** interfacing with ESC/POS thermal receipt printers and hardware barcode scanners.
- Configured automated build and release pipelines using **Fastlane & GitHub Actions**.

HAHALOLO SOCIAL TRAVEL NETWORK

05/2020 – 06/2025 (5 years)

Mobile Developer (iOS & Flutter) — Ecosystem of 6 Apps: Social, OTT, Travel, Fintech & Outsource

Tech Stack: iOS Swift, Autolayout, VIPER, MVVM, RxSwift, Starscream, Alamofire, IG List, SQLite, CallKit, Flutter, Add-to-App, Clean Architecture, Riverpod, Firebase.

- **[Hahalolo — Travel Social Network]** Developed flagship travel social app (Tour, Flight, Hotel, Handnotes) achieving **1M+ downloads**. Built native iOS modules (Swift, Autolayout) with **VIPER** and **Cocoapods**. Integrated Maps, Deep Linking, Autoplay Video, Firebase Crashlytics. Added Flutter via **Add-to-App**. Mobile team: 20.
- **[Halome — OTT Realtime Messaging]** Engineered real-time messaging app (SSO, Anonymous) using Native iOS Swift (**MVVM**). Integrated **Starscream (WebSocket)**, **Alamofire**, **IG List**, **RxSwift**, and **SQLite** offline. Implemented **CallKit** (Audio/Video), DeepLink, multimedia/location sharing. iOS team: 5.
- **[Haloki — Fintech]** Maintained & bug-fixed international money transfer app (**192 countries**) in Flutter. Ensured stable financial transaction flows, **KYC** verification and data security.
- **[Hahalolo Invest — Fintech]** Maintained & bug-fixed stock trading app in Flutter. Real-time chart rendering, portfolio management and buy/sell order workflows.
- **[Aquannets — Outsource for Galactic Holdings]** Developed aquaculture farm management app in Flutter. **Clean Architecture + Riverpod**, SQLite offline-first at farming ponds. Mobile team: 2.
- **[SocTrip — QC & Testing]** Served as **QC**: end-to-end testing, test cases, regression testing and quality sign-off before each Store release.

EDUCATION & CREDENTIALS

ACADEMIC

FPT POLYTECHNIC COLLEGE
Associate Degree in Information Technology (Software Development Major)

English: Professional Working • **Vietnamese:** Native

ENGINEERING DISCIPLINE & METHODOLOGY

PRACTICES

Architecture & Quality: Feature-First Clean Architecture, Strict SOLID principles, Ponytail Mindset (Zero Bloat, shortest working diff).

Testing & Delivery: 100% Deterministic Testing (Domain, Data, Widget), Automated CI/CD (Fastlane, GitHub Actions, Bitrise).